Name:- Aaryan N. Raorane Roll No:- 5004 branch:- AIDS
batch:- A1 SAPID:- 60018250070 Div:- 5

Experiment No 6

Q1 Define view with its syntax

⇒ A view is a virtual table defined by a stored SELECT query. It does not hold data itself. Every time you query it, the database runs the underlying query on the base tables and shows the result. Views are used to simplify complex queries, hide sensitive columns, and give different users different slices of the same data.

Syntax:-

```
CREATE VIEW ViewName AS
SELECT column1, column2
FROM TableName
WHERE condition;
```

Example:-

```
CREATE VIEW CSstudents AS
SELECT RollNo, Name, Marks
FROM student
WHERE Dept = 'CS';
SELECT * FROM CSstudents;
```

Modify and remove:-

```
CREATE OR REPLACE VIEW CSstudents AS
SELECT RollNo, Name FROM student WHERE Dept = 'CS';
DROP VIEW CSstudents;
```

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Q2 Write application of using triggers with syntax.

=> A trigger is a stored program that the database runs automatically when an INSERT, UPDATE or DELETE happens on a table. You never call it directly.

Syntax:

```
CREATE TRIGGER TriggerName
{BEFORE | AFTER} {INSERT | UPDATE | DELETE}
ON TableName
FOR EACH ROW
BEGIN
    statements;
END;
```

Inside the body, OLD holds the row values before the change and NEW holds the values after it.

Common applications:

Audit logging of changes, validating or correcting data before it is saved, keeping related tables in sync, and enforcing business rules that a simple constraint cannot express.

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Application1: Audit log of Salary changes

```
CREATE TABLE Salarylog (
    EmpId INT,
    OldSalary DECIMAL(10,2),
    NewSalary DECIMAL(10,2),
    ChangedOn DATETIME
);

DELIMITER //
CREATE TRIGGER logSalaryChange
AFTER UPDATE ON Employee
FOR EACH ROW
BEGIN
    INSERT INTO Salarylog
    Values (OLD.EmpID, OLD.Salary, NewSalary, NOW());
END //
```

whenever someone updates an employee's salary, a record of the old and new values is stored automatically.

Application2: Reduce stock when an order is placed.

```
DELIMITER //
CREATE TRIGGER ReduceStock
AFTER INSERT ON Orders
FOR EACH ROW
BEGIN
    UPDATE Product
```

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```
    SET Stock = Stock - NEW.Quantity
    WHERE ProductID = NEW.ProductID
END//
```

Inserting a row into Orders now updates the Product stock without any extra code in the application.

Remove a trigger:

```
DROP TRIGGER LogSalaryChange;
```

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